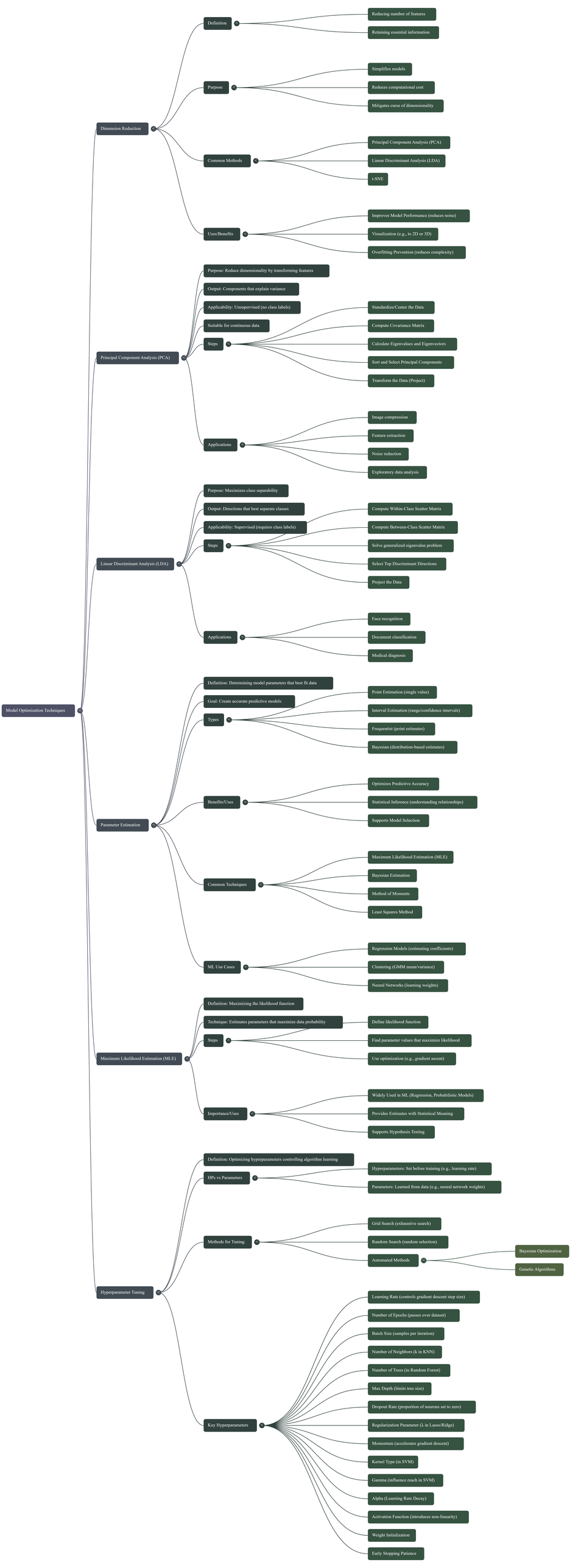
Model Optimization Techniques
Dimension Reduction
Definition
Reducing number of features
Retaining essential information
Purpose
Simplifies models
Reduces computational cost
Mitigates curse of dimensionality
Common Methods
Principal Component Analysis (PCA)
Linear Discriminant Analysis (LDA)
t-SNE
Uses/Benefits
Improves Model Performance (reduces noise)
Visualization (e.g., to 2D or 3D)
Overfitting Prevention (reduces complexity)
Principal Component Analysis (PCA)
Purpose: Reduce dimensionality by transforming features
Output: Components that explain variance
Applicability: Unsupervised (no class labels)
Suitable for continuous data
Steps
Standardize/Center the Data
Compute Covariance Matrix
Calculate Eigenvalues and Eigenvectors
Sort and Select Principal Components
Transform the Data (Project)
Applications
Image compression
Feature extraction
Noise reduction
Exploratory data analysis
Linear Discriminant Analysis (LDA)
Purpose: Maximizes class separability
Output: Directions that best separate classes
Applicability: Supervised (requires class labels)
Steps
Compute Within-Class Scatter Matrix
Compute Between-Class Scatter Matrix
Solve generalized eigenvalue problem
Select Top Discriminant Directions
Project the Data
Applications
Face recognition
Document classification
Medical diagnosis
Parameter Estimation
Definition: Determining model parameters that best fit data
Goal: Create accurate predictive models
Types
Point Estimation (single value)
Interval Estimation (range/confidence intervals)
Frequentist (point estimates)
Bayesian (distribution-based estimates)
Benefits/Uses
Optimizes Predictive Accuracy
Statistical Inference (understanding relationships)
Supports Model Selection
Common Techniques
Maximum Likelihood Estimation (MLE)
Bayesian Estimation
Method of Moments
Least Squares Method
ML Use Cases
Regression Models (estimating coefficients)
Clustering (GMM mean/variance)
Neural Networks (learning weights)
Maximum Likelihood Estimation (MLE)
Definition: Maximizing the likelihood function
Technique: Estimates parameters that maximize data probability
Steps
Define likelihood function
Find parameter values that maximize likelihood
Use optimization (e.g., gradient ascent)
Importance/Uses
Widely Used in ML (Regression, Probabilistic Models)
Provides Estimates with Statistical Meaning
Supports Hypothesis Testing
Hyperparameter Tuning
Definition: Optimizing hyperparameters controlling algorithm learning
HPs vs Parameters
Hyperparameters: Set before training (e.g., learning rate)
Parameters: Learned from data (e.g., neural network weights)
Methods for Tuning
Grid Search (exhaustive search)
Random Search (random selection)
Automated Methods
Bayesian Optimization
Genetic Algorithms
Key Hyperparameters
Learning Rate (controls gradient descent step size)
Number of Epochs (passes over dataset)
Batch Size (samples per iteration)
Number of Neighbors (k in KNN)
Number of Trees (in Random Forest)
Max Depth (limits tree size)
Dropout Rate (proportion of neurons set to zero)
Regularization Parameter (λ in Lasso/Ridge)
Momentum (accelerates gradient descent)
Kernel Type (in SVM)
Gamma (influence reach in SVM)
Alpha (Learning Rate Decay)
Activation Function (introduces non-linearity)
Weight Initialization
Early Stopping Patience